

nanochat baseline: decision brief

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Companion to the long report *nanochat: state of the project, October 2026* ([nanochat-state-2026-10.pdf](#)), referred to below as **LR**. Every number here comes from the LR. "Project" means [karpathy/nanochat](#) at HEAD [92d63d4](#) (2026-07-03), plus its leaderboard and discussions. Prices assume ~\$24/h for an 8×H100 node, the README's implied rate.

1. Which architectures are best

(a) Canonical dense recipe at HEAD: d24 + fp8, which is what `runs/speedrun.sh` trains

- **Recipe:** `--depth=24 --target-param-data-ratio=8 --fp8`, 1,384,122,122 params (of which 604.0M are value-embedding lookups; see the confirmed table below), ClimbMix-400B data, a 32k tokenizer trained in-run. It is base → SFT; RL is not part of the speedrun.
- **Delta vs the launch d20:**
 - deeper (d24 vs d20) and trained on ClimbMix instead of FineWeb-EDU;
 - attention: FA3 with a "SSSL" sliding window;
 - value embeddings, per-layer resid/x0 scalars, smear/backout, padded vocab;
 - a single MuonAdamW optimiser, fp8 matmuls, and no midtraining stage.
 - The MLP is still ReLU² (4× expansion, two matrices, no gate, no bias).
- **Score:** the record (leaderboard Run 6, [a825e63](#), Mar 14 2026) is CORE **0.2626**, val_bpb 0.71800, with 1.65 h of pretraining. The GPT-2 bar is 0.256525.

(b) Larger depth tiers that exist as reported runs

- **d26.** The README says only that "GPT-2 capability ≈ depth 26". Two FineWeb-era leaderboard entries used d26:
 - Run 2 (d26 + fp8, Feb 2): CORE 0.2578;
 - Run 3 (d26, 1M batch, Feb 5): CORE 0.2602.
 - There is no standalone d26 tier or price (the "\$300 d26" tier was not found, per LR Revalidation).
- **d32**, Oct 2025, official, released as SFT only. Same launch architecture as d20, just scaled: 1.88 B params, 37.6 B tokens. Base CORE **0.3168**; GSM8K 0.1994 after RL; ChatCORE 0.2734.
- **d34**, Nov 2025, official base. Launch-era architecture, 2.22 B params, trained at ratio 40 ("2× longtrained" vs Chinchilla). CORE **0.3382**, val_bpb 0.7045.

(c) Leaderboard and community variants that match or beat the baseline

- **#481 "\$73 / 3.04 h GPT-2"** (= leaderboard Run 1, d24, Jan 29 2026). This is Karpathy's own entry, not a community one. It used the pre-ClimbMix FineWeb-EDU data and predates smear/backout. CORE 0.2585. It has since been superseded by Runs 2–6 at the same task.
- **Run 5** (d24, ratio 8.7, 1.80 h): CORE **0.2690**, the highest d24 CORE on the board. It was superseded only because the leaderboard ranks time-to-0.2565, not CORE.
- **decoderstack-d24** ([ChrisMcCormick](#), community):
 - the same d24 shape, but trained with a different, non-nanochat training stack;
 - 5.84 B ClimbMix tokens, its own 32k tokenizer (it does not match other d24 tokenizers), CORE **0.2517**.
- **varlen-gsm8k** is the nanochat side branch, at [d9dffe9](#), into which decoderstack-d24 is verified to load. It is not an architecture of its own. Its name suggests variable-length attention plus GSM8K changes (UNCERTAIN: the branch contents were not read).

(d) Not reusable today

- **d34 and d32 official checkpoints.** They fail `strict=True` loading on master: they lack value embeddings, `ve_gate`, smear and backout, and their tokenizers were trained on FineWeb-EDU. They run only at "an equally old commit" (@svlandeg, #481).
- **RTX-4070-fork checkpoints** (Marcolini/nanochat-d24-base-*):
 - 910 M params, 9 heads, FineWeb-EDU, weights only, CORE 0.149–0.151;
 - they need that fork's GPT class.
- **Launch d20, Sofie d14, Nekochu d24, jkminder d26:** never released by Karpathy (d20), or pre-smear/backout, non-upstream format, or HF ports with the speedrun mechanisms ablated (LR §A.2).

Judgement

- **By CORE per dollar (our criterion), the best is the canonical d24 speedrun at HEAD.** It reaches CORE 0.2626 for ~\$48, about 0.0055 CORE/\$. For comparison, d32 gives ~0.0004 CORE/\$ (0.3168 / \$800) and d34 ~0.00014 CORE/\$ (0.3382 / \$2,500). It is also the only top entry that is runnable at HEAD.
- **By absolute CORE, the best is d34 (0.3382).** But it is frozen at a Nov-2025 commit on different data, so it is a reference point, not a usable architecture.
- **Among the d24 variants, the differences are within noise.** Run 5 (0.2690), Run 6 (0.2626) and decoderstack (0.2517) sit inside CORE noise of about ± 0.008 per run, with a 7-run spread of 0.0165 (LR §2.3). None of them beats the canonical recipe in a way that matters.

Sources: LR Executive summary, §A.2, §B.1, §B.3, §1.4, §2.3, Revalidation.

Confirmed d24 architecture and parameter budget (read from code at 92d63d4)

Item	Value at d24	Code
Width	<code>n_embd</code> = 24×64 = 1536 (depth $\times$ aspect_ratio, rounded up to a multiple of head_dim)	base_train.py L129–143
Heads	12 heads $\times$ head_dim 128; <code>n_kv_head</code> = 12 (GQA supported but unused)	base_train.py L133–138
Vocab	32,768 (<code>tok_train</code> default); pad-to-64 is a no-op here; wte and lm_head untied	tok_train.py L19 , gpt.py L170–177
MLP	<code>c_fc</code> 1536→6144 (4 $\times$ n_embd), <code>relu(x).square()</code> , <code>c_proj</code> 6144→1536. Two matrices, no gate, no bias. 18,874,368 params per block (8d ²). RMSNorm is parameter-free	gpt.py L131–141
Attention	q/k/v/o, no bias: 9,437,184 per block (4d ²). Window pattern SSSL, short window = 512 (the inline comment's "2048 -> 768" is stale), layers 3,7,...,23 full 2048	gpt.py L77–80 , gpt.py L298–313

Param group	Count
FFN, 24 blocks	452,984,832
Attention q/k/v/o, 24 blocks	226,492,416
<code>ve_gate</code> , 12 $\times$ (12→12)	1,728
wte	50,331,648
lm_head	50,331,648
Value embeddings, 12 odd layers $\times$ 32,768 $\times$ 1536	603,979,776
Scalars (resid/ $\times 0$ 48, smear 25, backout 1)	74
Total (asserted by <code>num_scaling_params</code> , gpt.py L390–409)	1,384,122,122

- **The ratio-8 denominator** is `transformer_matrices` + `lm_head` = **729,810,624**; wte and the value embeddings are excluded ([base_train.py L263–269](#)). $8 \times$ that $\rightarrow$ batch $2^{20} \rightarrow$ **5,568 steps** = 5,838,471,168 tokens ([L276–351](#)), i.e. 4.22 tokens per *total* param.
- **FLOPs** ([estimate_flops](#)). Projections are exactly $\frac{2}{3}$ FFN. With attention scores counted, the FFN share is 63.2% in the 512-window layers, 54.5% in the full layers, and 56.9% of the 4.775e9 training FLOPs per token. Attention projections are 28.5%, scores 8.3%, lm_head 6.3%.

2. Cost to reproduce each

Variant	Depth / params	Tokens	Wall-clock	\$	CORE	Hardware	Loads into HEAD?
Speedrun, current record (Run 6)	d24 / 1.384 B	5.84 B (5,568 steps $\times 2^{20}$; computed from HEAD code)	1.65 h pretrain; ~ 2 h full pipeline (2-2.5 h with setup, UNCERTAIN)	$\sim \\$48$; $\sim \$15$ on spot	0.2626 (averaging not stated)	8 $\times$ H100, fp8, FA3	Yes ; it is HEAD
Run 5	d24 / ~ 1.38 B	ratio 8.7	1.80 h	$\sim \$43$ (derived)	0.2690	8 $\times$ H100	Run with HEAD code (not a checkpoint)
#481 / Run 1 (FineWeb-EDU)	d24 / 1,384,124,976	not stated (4.33e19 FLOPs)	3.04 h	$\sim \$73$	0.2585	8 $\times$ H100	No; Jan-2026 commit and data
Runs 2-3 (FineWeb-EDU)	d26 / —	—	2.91 h / 2.76 h	$\sim \$70$ / $\sim \$66$ (derived)	0.2578 / 0.2602	8 $\times$ H100	No; Feb-2026 commits
Launch d20 (#1, Oct 2025)	d20 / 560,988,160	11.22 B	3 h 51 min	\$92.40	0.2219	8 $\times$ H100	No; never released, and old architecture
d32 (official, SFT only; #8)	d32 / 1,879,048,192	37.58 B	~ 31.3 h	$\sim \$800$	0.3168 (base)	8 $\times$ H100	No ; needs an Oct-2025 commit
d34 (official base)	d34 / 2,217,082,880	88.7 B (ratio 40)	~ 100 h	$\sim \$2,500$	0.3382	8 $\times$ H100	No ; "equally old commit"
decoderstack-d24 (community)	d24 / not stated	5.84 B	not stated in LR	not stated in LR	0.2517	training hardware not stated; load verified on an H100	UNCERTAIN : verified only on varlen-gsm8k @ d9dffe9, not on master
Marcolini d24 (RTX-4070 fork)	"d24" / 910 M	—	—	—	0.149-0.151	RTX 4070 (fork)	No; fork's GPT class, no tokenizer

Notes on the table:

- \$ marked *derived* is wall-clock × \$24/h, our arithmetic, not quoted by upstream.
- CORE noise is about ± 0.008 per run, so differences under ~ 0.01 are not real.
- val_bpb is **not comparable** across the Mar 4 ClimbMix switch, nor across the Jul 3 token_bytes fix (2ce972a). CORE is the cross-era metric.
- No current-speedrun report card (chat metrics) is published (UNCERTAIN).

Sources: LR §B.1 (tiers and leaderboard row), §B.2 (launch d20), §A.2 (checkpoints, decoderstack, Marcolini), §2.3 (noise, baselines), Revalidation.

3. Which baseline our paper should cite

Recommendation: cite and re-run the canonical nanochat d24 speedrun, pinned to commit 92d63d4. Report it as the baseline both as Karpathy's published number (CORE 0.2626, Run 6) and as our own reproduction, mean $\pm$ sd over 3 seeds. Cite d32/d34 only as reported numbers, and use no community variant as a baseline.

Why this one

- **Reproducible by reviewers:** one script, standard 8×H100 at $\sim$ \$48 ($\sim$ \$15 on spot), or one H100 (see below). It is the only top configuration a third party can re-run at HEAD; every alternative needs an old commit, another training stack or a fork.
- **Canonical:** the README and leaderboard define it as "the" nanochat model, so "nanochat d24 speedrun @ 92d63d4" is an unambiguous citation.
- **Cheap enough to seed:** CORE noise (± 0.008 per run) is about the size of our effects, so we need our own multi-seed mean.

The baseline is a moving target: what the paper must pin

The repo has been quiet since 2026-07-03, but it switched datasets on Mar 4 2026 and changed the val_bpb byte accounting on Jul 3 2026, so a "nanochat baseline" without these pins is meaningless:

- **Code:** commit 92d63d4, the exact speedrun.sh flags (`--depth=24 --target-param-data-ratio=8 --fp8`), plus `torch==2.9.1+cu128` (fp8 is nanochat's own fp8.py; no torchao), and the FA3 kernel revision fetched from the Kernels Hub.
- **Data:** karpathy/climbmix-400b-shuffle, the 170 downloaded shards, with val = the last shard (shard_06542). Also the sha256 of our trained tokenizer, since it is trained in-run.
- **Token budget:** ratio 8 gives $5,568 \text{ steps} \times 1,048,576 = 5.84 \text{ B tokens}$ at HEAD (LR Part 3). Record it from our log anyway.
- **Eval:**
 - CORE = the 22-task DCLM bundle from S3; store its sha256 (size unverified).
 - val_bpb as computed after 2ce972a.
 - ChatCORE is 5 tasks at HEAD (it was 6 at earlier commits); say which.
- **Hardware:** 8×H100 with fp8 and FA3. Wall-clock claims hold only on this setup. Our 5090 runs fall back to SDPA, with no FA3.

Should we run d32/d34 as an upper reference?

No; cite them.

- Re-running costs $\sim$ \$800 for d32 or $\sim$ \$2,500 for d34, and at HEAD it would produce a *new* model, not Karpathy's. Running the published checkpoints would mean the old commit and FineWeb-EDU data, not comparable with our ClimbMix arm.
- Their reported CORE values (0.3168, 0.3382) are enough to show where the scaling curve goes, framed as "launch-era architecture, FineWeb-EDU, as reported".
- Only buy a stronger tier if a reviewer asks whether LUT gains hold at scale. Even then, a HEAD d26 or d32 run on our recipe is the right experiment, not the old checkpoints.

The risk of citing community and leaderboard variants

- decoderstack-d24 has a different training framework and tokenizer, loads only on a side branch, and scores 0.2517, below the GPT-2 bar and within noise. Runs 1–3 are FineWeb-era and

superseded; the RTX-4070 fork is a different model class. They are fine as related-work mentions, but never as the baseline row.

One caveat on the comparison itself

LUT layers are not `nn.Linear`, so the LUT arm cannot use fp8 in its FFN (LR §2.4). The headline baseline row stays the canonical fp8 run, because that is what is citable. But the **matched control** for the LUT claim should be the same d24 at the same commit with `--fp8 off`, so the comparison isolates the FFN rather than the precision.

Can one H100 80GB replace 8×H100? Yes, with caveats (LR Part 3)

- **Numerically comparable, yes.** At HEAD, the global batch (2^{20} tokens), the step count (5,568), the LRs and the weight decay all derive from param count and token ratio, never from GPU count. Only grad-accum changes: $4 \rightarrow 32$ at dbs 16.
- `MuonAdamW` runs single-rank natively. Its maths is per matrix, so sharding only decides who computes what.
- The run is not bit-identical: rank-strided data order and fp32 reduction order differ. But the same is true of two 8-GPU runs. Expect CORE within noise (± 0.008); the README says “~identical results”.
- CORE eval is rank-agnostic. Report the standalone `base_eval` CORE. The in-training CORE is a 500-example-per-task subsample and is not the reported number.
- **Caveats:**
 - edit `--nproc_per_node=8` in `speedrun.sh`;
 - memory is $\approx 70\text{--}75$ GB at dbs 16 (UNCERTAIN; extrapolated from one 5090 report). Fall back to dbs 8, which has the same maths.
 - It takes $\approx 15\text{--}18$ h end to end at about the same GPU-hours, so about the same \$ ($\approx \$45\text{--}55$ at $\$3/\text{GPU-h}$).
 - On spot, add `--save-every`. Resume works but is only approximate for the dataloader.
- **Upstream evidence:** #819 trained the same d24 (5,568 steps, $256\times$ accumulation) on one RTX 5090 in 41.4 h, reaching CORE 0.2702 with the non-standard `L` window.

What we should actually do

1. **Run** the canonical speedrun at 92d63d4 on rented 8×H100, **3 seeds**: $\sim 3 \times \$48 \approx \150 . Use spot pricing if available, $\sim 3 \times \$15$.
2. **Run** the matched dense control, d24 at 92d63d4 with `--fp8 off`, 3 seeds. Budget **~\$170**: fp8 gave $1.17\times$ throughput at d26 (LR Part 3 / LOG), so fp8-off is ~ 2.3 h $\approx \$56$ per run. This is our estimate.
3. **Pin and publish:**
 - the commit, flags and package versions;
 - the dataset name and shard range;
 - the tokenizer, eval-bundle and FA3 hashes;
 - the logged token count;
 - the hardware and seeds. Release our d24 checkpoint and logs, since no current-architecture base checkpoint is public.
4. **Cite, don't run:**
 - Run 6 (CORE 0.2626) as the canonical reference;
 - the GPT-2 bar (0.256525);
 - d32 (0.3168) and d34 (0.3382) as reported upper reference points;
 - the launch d20 (0.2219) only for history.
5. **Do small-scale LUT iteration at home first.** Run dense d12 at the same commit on the 5090: no rental cost, and no published d12 number exists, so we must make our own. Buy the 8×H100 runs only once the LUT arm works.
6. **Rough total: ~\$320 for the d24 baseline package**, about \$370 with a contingency rerun. Spot pricing, where available, cuts this roughly threefold.

Sources: LR Executive summary (recommendation), §A.2 (checkpoint loadability), §A.3 (FA3 fetch), §B.1, §B.3, §C (software stack), §2.3 (eval harness, noise, missing d12 numbers), §2.4 (fp8 only on `nn.Linear`; fork at 92d63d4).